

Date: 17.09.2024

Roll No. _____

III SEMESTER B.Tech (IT, ITNS)

MID-SEMESTER EXAMINATION 2024

Course Code : INECC 305/ITECC305 / ~~INECC06/ITECC06~~

Time: 1:30 Hrs

Course Title : Probability & Stochastic Processes

Max Marks: 20

Note: -For Re-registration ie INECC06 / EIECC05, marks shall be scaled up to 25 marks. Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
Q 1	Let A and B be events with probabilities P[A] and P[B] a. Find $P[A \cup B]$ if A and B are independent b. Find $P[A \cup B]$ if A and B are mutually exclusive. c. Show that $P[A \cap B \cap C] = P[A B \cap C]P[B C]P[C]$	03	CO1
b	Consider the two random variables (X,Y). Find the region of the plane corresponding to the events $A = \{X + Y \leq 10\}$, $B = \{\min(X,Y) \leq 5\}$, and $C = \{X^2 + Y^2 \leq 100\}$.	01	CO2
Q 2	Two friends decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., find the probability that the first to arrive has to wait longer than 10 minutes	03	CO1
b	The random variable X has CDF $F_X(x) = \begin{cases} 0 & x < -3 \\ 0.4 & -3 \leq x < 5 \\ 0.8 & 5 \leq x < 7 \\ 1 & x \geq 7 \end{cases}$ Write $P_X(x)$, the PMF of X.	01	CO2
Q 3	Let X be a random variable with pdf $f_X(x) = \begin{cases} 0 & x < 0 \\ ce^{-2x} & x \geq 0 \end{cases} \quad (c > 0)$ (a) Find c (b) Let $a > 0$, $x > 0$, Find $P[X \geq x + a]$ (c) Let $a > 0$, $x > 0$, Find $P[X \geq x + a X \geq a]$	03	CO2
b	Compute $E[X ^n]$, if X is Uniformly distributed with zero mean and variance equal to 2.	01	CO2
Q 4	The joint density function of X and Y is given by: $f_{X,Y}(x,y) = \begin{cases} 2e^{-x}e^{-y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$ Compute (i) $P\{X > 1, Y < 1\}$ (ii) $P\{X < Y\}$ (iii) $P\{X < a\}$ (iv) $f_X(x)$	03	CO2
b	What are the properties of Joint CDF, Joint PDF and Joint PMF.	01	CO1
Q 5	Suppose the joint PMF of a bivariate r.v. (X,Y) is given by: $p_{X,Y}(x_i, y_j) = \begin{cases} \frac{1}{3} & (0,10), (1,0), (2,1) \\ 0 & \text{otherwise} \end{cases}$ (i) Are X and Y independent? (ii) Are X and Y uncorrelated?	03	CO3
b	Random variable X has mean 'm' and variance σ^2 . Compute the mean of $Y = 2X = 2X^2 + 3X - 4$	01	CO3